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Checkpoint 1

Past bond programs were structured to recommend projects in each Ward to try to maximize the likelihood of a successful outcome in a voter election. Can the results be provided in a map format overlaid on the City Ward Map to make sure we are getting good representation for each Ward? This may require that we have to look at more than the top 20 locations in the City. In mapping these collisions/volumes/collision-volume aggregate, we would envision some sort of heat map for demonstration purposes.

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1. Problem Understanding

The objective is determine which sections of Norman need the most attention to improve voter turn out. I will achieve this objective by plotting the collisions/volumes/collision-volume aggregate on top of the City of Norman ward map. I'll use a tool like Seaborn to plot the data on the map.

The data would have to be grouped by its ward rather than Norman as a whole. For example, I would plot the top 25 most dangerous intersections for each ward and the top 25 most congestion roads for each ward.

2. Metrics Definition

- **Traffic volume:** Number of vehicles per day.
    - To determine typical traffic volume, I will calculate an average traffic volume over the course of the most recent Fall season. I am choosing the Fall season because the University of Oklahoma is in session, local schools are in session and football is in season. Holidays will be excluded to prevent the data from being skewed. If a road had a working day, this should be representative of what that would look like.
- **Volume-to-capacity ratio (V/C):** Traffic volume ÷ roadway capacity.
    - Explain how you would determine capacity using lane configuration and the provided ACOG guidelines, including which capacity column you would use.
- **Collision count:** Number of unique collisions assigned to a roadway segment during the selected period.
- **collision-volume aggregate**: collisions ÷ v/c ratio
    - I'll calculate a ratio of 'for each day' 'what was the volume-to-capacity-ratio' and how many collisions occurred that day?' Roads that have many collisions for a low road usage would be considered dangerous and problematic.

3. <u>Data & Columns Needed</u>

- **ACOG traffic count files** for the selected analysis period
- **ACOG Guidelines** for Roadway Capacity vs. Number of Lanes
- **Norman GIS map** and the **Ward Maps** layer

Information needed	**Columns or source**	**Purpose**
Location identifier	Loc ID or LOCAL_ID	Identify locations and match repeated observations
Roadway and segment endpoints	On/ON_ROAD, From/FROM, To/TO	Define the roadway segment
Traffic volume and date	Available count/date fields and text columns	Determine daily volume and observation period
Coordinates	Latitude and longitude fields	Locate traffic observations
Roadway classification	Functional Class, where available	Help identify project-eligible roads
Lane configuration	GIS and aerial imagery	Select the applicable capacity
Ward assignment	Ward Maps layer	Group segments geographically

Data preparation plan:

1. **Choose and document the analysis period.** Check actual count dates rather than assuming all observations match the fiscal year in the filename.
2. **Standardize column names and roadway names.** Match equivalent fields across files and resolve inconsistent naming.
3. **Prepare traffic counts.** Extract vehicle counts and dates from text, keeping speed values separate. Add opposing directional counts only when they belong to the same observation.
4. **Define consistent segments.** Match each count location to a roadway segment. Avoid adding counts from successive locations along the same road, which could count the same traffic more than once.
5. **Determine capacity.** Identify lane configuration, including divided roads or center turn lanes, and match it to the ACOG guidelines.
6. **Assign wards.** Use the GIS ward boundaries and document how segments crossing or following ward boundaries will be handled.

7. **Check data quality.** Flag missing counts, uncertain lane configurations, and unmatched locations. Exclude unresolved records from V/C calculations and report those exclusions.
8. **Aggregate Data**: group traffic information by ward before performing analytical calculations